

Automated Financial Data Pipeline: ETL Processing and Portfolio Analytics

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1. Project Overview

This project is an end-to-end ETL pipeline project built using Python and a public financial API to extract stock market data, process and transform it into analytical datasets, and visualize insights through interactive dashboards.

A financial firm needs to monitor the performance and risk exposure of a small technology stock portfolio. The objective is to transform raw market data into structured insights that support decision-making around portfolio performance, volatility, and diversification.

Using Python, daily stock market data is extracted from an external API, cleaned, and transformed into analytical datasets. Key financial metrics such as daily returns, rolling averages, and volatility are engineered to better understand both short-term fluctuations and longer-term trends. The processed data is then loaded into Microsoft Power BI to create interactive dashboards for performance tracking and risk analysis.

2. Problem statement

The financial firm needs a lightweight analytics system to:

- Track daily performance of technology stocks
- Understand volatility and risk exposure over time
- Replace manual reporting with a structured data pipeline

3. Skills and Tools demonstrated

- API integration and data extraction
- Python data processing (Pandas)
- ETL pipeline design
- Financial data analysis
- Dashboard development in Power BI
- Business insights

4. Data Source

The dataset was retrieved from the Alpha Vantage API, providing daily stock market data including:

- Open price
- High price
- Low price
- Close price
- Volume

The primary asset analyzed in this project is **AAPL**, along with a simulated portfolio structure for comparative analysis.

5. ETL Pipeline

5.1 Extract Phase

- Data is extracted from the Alpha Vantage API using Python
- JSON response is converted into a structured pandas DataFrame
- Daily time series data is isolated for analysis

5.2 Transform Phase

Data Cleaning

- Standardized column names
- Converted string values to numeric types
- Sorted data chronologically
- Removed duplicates
- Handled missing values using forward fill

Feature Engineering

The following metrics were created:

- **Daily Return:** percentage change in closing price
- **Rolling 7-Day Average:** smooths short-term fluctuations

- **Volatility:** measures variability and risk of returns
- **Weighted Return:** adjusts performance based on portfolio allocation

5.3 Load Phase

Two structured datasets were created:

- “stock_prices.csv”: time-series dataset for analysis
- “portfolio_summary.csv”: aggregated dataset for portfolio comparison

These datasets were loaded into Power BI for visual analytics.

6. Power BI Dashboard

- **Stock Price Trend (Line Chart):**
Displays the closing price over time to identify overall market direction and trend behavior.
- **Daily Returns (Bar Chart):**
Shows daily percentage gains and losses to evaluate short-term market fluctuations.
- **Volatility Analysis (Line Chart):**
Tracks risk levels over time by measuring variability in daily returns.
- **Rolling Average (Trend Indicator):**
Smooths price movement to highlight underlying trends and reduce market noise

AAPL Stock Performance & Risk Analytics

298,87

Latest Closing Price

0,00

Daily return average

0,01

Average of volatility

Stock Price Trend



Volatility by date



Daily Returns



Rolling Average

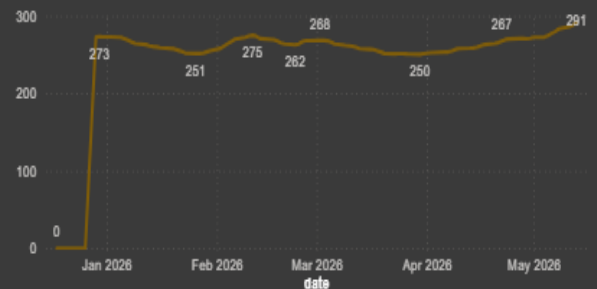


Figure 1: Power BI Dashboard: AAPL Stock Performance & Risk Analytics

7. Key insights

7.1 Stock Price Trend Insights

- AAPL opened the period at **\$272** and almost immediately began declining, hitting a low of **\$247** in mid-January
- After that low, the stock attempted a recovery up to **\$278** in early February, but couldn't hold it and fell back down to **\$256**
- The turning point came in late March/early April where the stock found a floor around **\$247–\$259** and stopped making new lows
- From April onwards, AAPL entered a clear and consistent upward trend, climbing all the way to **\$299 by mid-May**, its highest point in the entire dataset
- **Key insight:** The first 3 months were volatile and bearish, but the stock made a strong bullish recovery in the final stretch, ending the period up overall

7.2 Daily Returns Insights

- Most days were small, normal movements between -1% and +2%
- The worst day was **-5.0%** in mid-February
- The best single day was **+4.1%** in early February, which came shortly after the worst days
- From **April into May**, the balance clearly shifted, positive returns of +2.6%, +2.6%, and +3.2% started appearing more frequently with smaller and less frequent negative days
- **Key Insight:** The return pattern confirms Q1 was high-risk and unstable, while Q2 shows improving momentum with more consistent positive returns

7.3 Volatility Insights

- Rose sharply through January, reaching **0.020** by late January/early February, meaning the stock's daily swings were becoming increasingly unpredictable
- Volatility peaked at **0.027** in early March, marking the riskiest period in the entire dataset
- After that peak, volatility dropped significantly to a low of **0.007** in late March

- From April into May, volatility climbed again to **0.021**, which is notable because prices were also rising during this period
- **Key insight:** The firm faced the most risk in Feb–March, got a brief period of calm in late March, but should note that the current price rally in May still carries elevated volatility

7.4 Rolling 7-day average Insights

- the rolling average opened around **\$273**, but it declined to **\$251** in early February, confirming that the price weakness in chart 1
- From April it dipped briefly to **\$250**, its lowest stable point, before turning upward decisively
- By May the rolling average reached **\$291**, its highest point, confirming that the recent price rally is sustained
- **Key insight:** The rolling average is your most reliable trend confirmation tool, and it clearly shows that what's happening in May is a real, momentum-backed recovery, not a short-term spike

8. Conclusion

This project successfully delivered an end-to-end ETL pipeline that transformed raw API data into structured, actionable financial insights, replacing what would otherwise be a manual reporting process with a repeatable and scalable analytics system.

The analysis revealed that AAPL had a challenging first quarter, marked by notable price declines, large single-day losses, and peak volatility of 0.027 in early March 2026. However, from April onwards the stock staged a clear and sustained recovery, reaching a high of \$299 by mid-May, a trend confirmed across all four dashboard visuals. It is worth noting that while the recovery is real, volatility remains moderate at 0.021, meaning the gains still carry some risk.

Overall, this project demonstrates the value of a structured data pipeline in a financial context. The firm now has a systematic framework to monitor performance, quantify risk, and identify trends as they develop, laying the groundwork for more advanced analytics

such as multi-asset comparisons, benchmark tracking, and automated risk alerts in the future.